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A Quantum Approximate Optimization Algorithm Based on Blockchain Heuristic Approach for Scalable and Secure Smart Logistics Systems

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Abstract

Smart logistics and supply chain play can determine the success or failure of any business. The cost, time, and carbon footprint are critical elements to be considered. Smart logistics solely consume 53% of the company's income and produce up to 10% of its carbon footprint. Moreover, the time consumed in transportation and supply chains from the resource acquisition to the client contributes to the business profit. Enhancing smart logistics systems by selecting the optimal route is a hard problem even for today's supercomputers. On the other hand, Quantum-based processing and Quantum algorithms are proved to solve convoluted computation to attain a heuristic system swiftly compared with classical processing methods. Notably, Quantum approximate optimization algorithm (QAOA), as a variational Quantum algorithm for approximately solving discrete combinatorial optimization problems can be deployed into the smart logistics dilemma to improve the scalability of the system, decrease the time, thus reducing the carbon footprint and smart manufacturing system cost. Moreover, blockchain, as a secure distributed ledger, is capable of bringing the desired security to the smart logistic system. To this end, we propose an improved QAOA based on blockchain technology to improve the scalability and reduce the cost of smart logistics.

Keywords

Quantum Approximate Optimization Algorithm, Blockchain, Smart Logistics, Supply Chain, Smart Transportation

1. Introduction

Logistics and supply chains are an inseparable part of any business that engenders the process of resource acquisition, storage, and delivery to the end client. It is reported that in 2017, major companies spent above 1.5 trillion dollars on logistics costs [1]. In the United States alone, logistics and supply chains consume 7.7% of national GDP per year. The cost of logistics is increasing over years due to many factors including fuel price, distance, and the growing demand for delivered products from the end client. Large companies such as Amazon report an increasement of 2 billion dollars per quarter spent on logistics in 2020 [2], the company CEO stated that the cause of this loss is due to the transportation costs. Logistics

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are a complex dilemma for businesses around the globe as they cost companies a large percentage of their return. Moreover, the logistics chain contributes to the carbon emission by 10% due to the heavy transportation method deployed including trucks, ships, trains, and airplanes. To reduce the cost and carbon footprint emission, a more optimized and modern approach shall be taken. Optimizing logistics systems is considered to be one of the most complex problems to solve using today's supercomputer. The problem with the current logistics system is not only an economic and environmental issue but also expanding to create a security and privacy dilemma. Modern logistic systems use classical satellite-based communications systems which are prone to cyber-attacks. 2020 marked a year of active attacks on logistics and transportation companies. Most of the attacks were in form of ransomware, and all of them used the communication channel deployed between vehicles and the company headquarter [3]. Fast and secure communication between the supply chain's tiers is the main pillar to a successful smart logistic system [4] and securing it against any potential cyber-attack is critical. Optimizing smart logistics systems and preserving the security and privacy required level could be a complex a challenging task using today's system [5–8].

On the other hand, Quantum information science can be described as the future era of computation [9, 10]. Based on Quantum mechanics laws, a Quantum computer deploys the states of elementary particles, which is known as the spin or the internal angular momentum, to create the Quantum bits (Qubits). A Qubit holds both values, 1 and 0, at the same instance; 1 according to the spin-up, while 0 refers to spin-down. This functionality enables Quantum computer with n Qubits to perform 2^n operation parallelly, which increases the computational speed exponentially. Quantum computers and Quantum-based algorithms can be deployed to computer complex problems in a short amount of time and produce desirable results. Since smart logistics deal with a large amount of data, optimizing the system will require the development of a Quantum-based algorithm. To this end, we propose in this paper the deployment of Quantum approximate optimization algorithm (QAOP) with bockchain to optimize the smart logistics system all while preserving the required security and privacy using blockchain technology. QAOP is deployed to find the ground state of an Ising Hamiltonian, which solves most NP-complete problems [11]. The benefits of using QAOA lies in its hybrid algorithm nature that can be executed on both classical and Quantum computers, the Quantum part of the algorithm can be implemented using a Quantum circuit with p level, where the higher the p level the better approximation results will be, thus promising higher performance for optimization problems [12, 13]. In our proposal, QAOP is used to solve the optimization problem for smart logistics systems by calculating and selecting the heuristic direction for logistics transportation to follow from the resource collection to the customer delivery. Moreover, a private Blockchain network is used to secure and preserve the privacy of communicated data between smart logistics different tiers by creating a trustworthy cluster.

The following points summarize our main research contribution:

- We propose a QAOP to reach heuristic optimization by deploying it to calculate and optimize the travel path to select the most optimized one to follow to reduce cost, carbon footprint, and time.
- A private blockchain network is deployed as well in the proposed framework to maintain the required level of private communication and secure data sharing between different tiers of the smart logistics system.
- The proposed solution is simulated under IBM Quantum Lab and shows promising results of cost, carbon footprint, and time reduction.

The rest of the paper is organized as follows: Section 2 depict the related works and explain the main technologies and algorithm used in this framework. The main proposed idea is described in Section 3 with a detailed explanation of the process flow including blockchain-based communication phases, max-cut problem formulation, and the QAOA-based solution. Section 4 presents the experimental results of the proposed framework. And we conclude this work with Section 5.

2. Related Work

The complex nature of smart logistics and smart transportation makes it a difficult task for classical computers to optimize. Security and privacy are also other dilemmas that face smart logistics systems. Quantum-based algorithms are one of the solutions that can be deployed for the optimization problem, and Blockchain as well as one of the most known keys for security and privacy. In the following, we will discuss some other proposed solutions in multiple states of art.

2.1 Seminar Contribution

Woschank et al. [14] presented a review of further directions for artificial intelligence, machine learning, and deep learning that can be applied in smart logistics. The paper discusses the deployment of multiple technologies to optimize smart logistics, and based on their systematic study, the results collected were put into fruitful implications for future research in cutting-edge technologies for smart logistics. Khatib and Barco [15] took another approach to optimize the smart logistics and supply chain systems. The authors focused on the connectivity aspect of smart logistics where they proposed a system to exploit the application-specific optimization capabilities of 5G networks. The authors argued on the necessity of improving the connectivity network to optimize the system in general by using Fuzzy logic over the requirements of each application and a bid data prediction module to predict the traffic and divide the resources. Tang [16] proposed a novel scheme of functions for intelligent analysis, perception, optimization of decision-making based on exhaustive research on the current status of information technologies in the logistic system to create sustainable optimized smart logistics. Through the proposal analysis, the system shows to have good implementation effects. On the other hand, Arumugam et al. [17] deployed the advancement of Internet-of-Things (IoT) networks to automate and increase real-time decision-making in supply chain management and logistics. Moreover, the authors made use of smart contracts to assure trust between different tiers of the system. The proposed solution includes smart contracts, a logistics planner, and condition monitoring of the assets in a smart logistic system. The results prove the accountability, traceability, and liability for asset handling in smart logistics. Bartolacci et al. [18] studied multiple optimization methods and techniques to be used for logistics such as strategic, tactical, operational, and collaborative techniques. The authors stated that the key success to logistics optimization is the knowledge of the process and tiers to be modeled by the analysts and managers. While Said and El-Rayes [19] proposed an automated multi-objective constructor's logistics optimization system (AMCLOS) to plan an optimal material supply and storage. The proposed method is proved to provide useful insights on the supply chain to reach an optimal system.

Optimization of smart logistics, especially smart transportation in a supply chain will benefit businesses, governments, the environment, and end-client by reducing the costs, charges, time, and carbon footprints. Using the advancement of today's hardware and software, smart logistics is indeed improving, but not to the desired level of costs cut down. As discussed in the above papers and related works, various methods have been used in literature to improve smart logistic systems such as machine learning, communication networks' optimization, and blockchain. However, the results proposed do not consider all the desired requirements. Moreover, security issues were not taken into consideration as well. To this end, and to improve and fasten the optimization phases, we believe that the deployment of Quantum-based algorithms and blockchain is the solution that can cover all the aspects of this dilemma. The Quantum-based algorithm can be used for fast and efficient problem computation to reach heuristic optimization, while blockchain can be deployed to enhance security and privacy.

2.2 Key Considerations

The primary considerations of the proposed architecture are depicted as follows:

- **Optimization:** Smart logistics and smart transport systems combined advanced IoT devices, sensors, and big complex data, including communication between multiple tiers of the smart supply chain. The end goal of this application is to improve the quality of service (QoS) and quality of experience (QoE) for the business, suppliers, as well as the end client. Smart logistics are growing fast and become more and more complicated as more tiers and end clients are contently added by the time. Thus, creating a scalable infrastructure that enables healthcare providers to maintain the needs of their fast-growing components requires fast and efficient computation and optimization methods that can be provided by the deployment of Quantum-based algorithms.
- **Efficiency:** The efficiency of computations is another critical requirement for smart logistics data. However, with the existing and fast-growing large-scale applications, efficiency is not always guaranteed as the needs and demands are higher than the current classical services and systems. Therefore, a fast and efficient computation system is highly required to balance the scalability and efficiency of smart logistics processing, thus improving the QoS, and reducing the cost.
- **Security and privacy:** Smart logistics, supply chains, and smart transportation systems are designed to collect numerous amounts of data, notably real-time acquired data, and information. Companies' private information such as product ID, files, locations, companies' status, and many more can lack to hackers intentionally and unintentionally. Thus, securing this data is a critical requirement. This issue is addressed by giving access to only pre-authenticated users.
- **Confidentiality:** The main challenges of Smart logistics, supply chains, and smart transportation are the security and privacy of information. Sensitive data are vulnerable to attacks from third parties, which may lead to data manipulation, loss, or exposure by unauthorized individuals. In addition, smart logistics, supply chains, and smart transportation are more prone to security attacks and disclosure of their data as they hold sensitive information about the company.
- **Integrity:** Data manipulation or modification by any unauthorized party is another critical issue in smart logistics, supply chains, and smart transportation environments. An attacker may use malware, ransomware, or masquerade attacks to erase the data.
- **Availability:** In a smart logistics, supply chains, and smart transportation environment, due to its sensitive nature, data should be accessible by authorized individuals and parties whenever needed. Assuring data availability is critical to the functionality of smart logistics systems as it relies mainly on data acquisition.

3. System Model

3.1 Proposed System Overview

The proposed framework consists of five main tiers: the raw material supplier, warehouse, manufacturing, retailers, and end-clients (Fig. 1). A supplier engenders several supply chains [20]. The supplier is in charge of providing raw materials for manufacturing, the materials are usually not proceeded and in their original form. On the other hand, warehouses are the main pillar of every successful business model globally [21]. A warehouse is the storage space where raw materials from suppliers can be stored before the processing, as well as after the processing phase. The location of warehouses is critical as it determines the time and distance to deliver raw material and finalized products. Manufacturers are where the raw materials are turned into physical and usable products [22]. Businesses either own the manufacture or use third-party manufacturing facilities. Smart manufactories engender various industrial IoT devices and numerous sensors that help keep the production line active. Retailers are the first buyers of the physical product; the final version of the products is usually sold to retailers so they can reach end-costumers easier [23]. Retailers may be offline retailers with physical stores or online retailers with a website or application the posts the products on for the clients. As a final step for this chain resides the

end-customer, the client is usually the direct consumer of the products bought from retailers [24]. Since the client is the major contribution to the success of a business, bringing products closer to him and faster can directly affect and impact the profit of the company.

Apart from the classical supply chain, our proposal for smart, effective, and secure logistics includes the deployment of the Quantum approximate optimization approach and blockchain technology. Zhou et al. [25] define QAOA as a hybrid Quantum-classical variational algorithm dedicated to bringing a heuristic optimization to a complex problem. Moreover, in this proposal, private blockchain can be deployed to enhance the privacy and security of the communication [26, 27] between different tiers of smart logistics by authorizing and authenticating the participated entities in a communication channel. Using this method, the head office can securely collect the data from other tiers and compute using the QAOA the heuristic optimized path for the transportation to take, thus reducing the time, carbon emission, and cost of shipping.

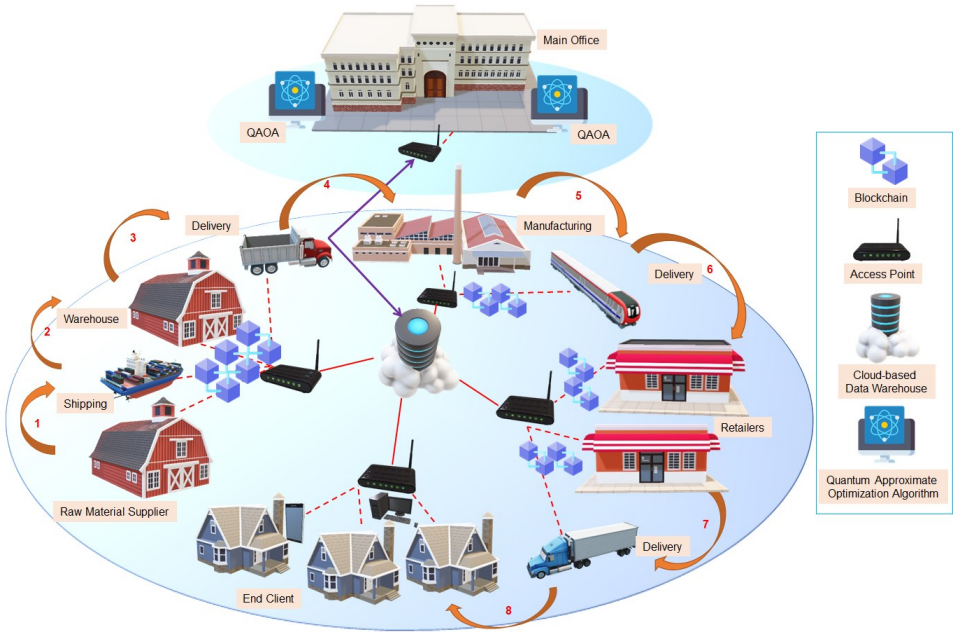


Fig. 1. Overview of the proposed system.

3.2 Blockchain-based Secure Communication

Due to the critical information communicated between different components of smart factory and logistics such as products identification and inner company data, securing the transmitted messages is an indispensable phase. To this end, we propose in this paper the deployment of private blockchain to ensure that only the authenticated parties have the right to participate in the communication and view the messages and the delegated data. The blockchain-based secure communication can be divided into two main phases: registration phase and communication phase. The details of each phase can be depicted in the following subsection.

3.2.1 Registration phase

Each factory has a network of entities they deal with, they can be between the same company such as warehouses and delivery trucks, or outside the company such as the supplier they deal with or third-party delivery services. The communication between these entities needs to be secured against any data leakage or cyber-attacks discussed previously. In this paper, we consider that the company owns a private

Blockchain network controlled by the main office, then the main office will validate the registration of other entities they deal with first before establishing the secure communication channel. Each entity is required to register themselves in the private blockchain ledger before initializing the communication. Algorithm 1 depict the authentication phase.

Algorithm 1. Registration phase

1: **Input:** Entity identification and public key
2: **Output:** Private authenticated network
3: **Process:**
4: ET_n.Send(<ET_{id}, Msg, *t, N>, request, MO);*
//with ET_n.: Entity identification, t: *timestamp, MO*: *Main Office*
5: MO.Verify (<ET_{id}, Msg, t, N); *//N: the Msg nonce*
6: MO.*Prepare(<v, N, d>, Msg); //d: the Msg digest, v: view* identity
7: MO.*Broadcast(<v, N, d, s>, ET_n); //s: the digest signature* of MO
8: **for** *i = 1 to n*
{
9: ET_i.*Receive(<v, N, d, s >);*
10: ET_i.*Verify(<v, N, d, s >, f, n);*
11: ET_i.*Prepare(<v, N, d, s >);*
12: ET_i.*Broadcast(<v, N, d, s >, $ET_{|n-i|}$);*
13: ET_i.*count(<f: fault d>, count m);*
}
14: ***while*** count: m>(2f+1) ***then***
{
15: MO.Compute (<*v, N, d, s* >, *Msg*, *Si*); // *with Si: The Special identification*
}
16: **for** k=1 to n
{
17: MO . *finalCheck*(BSk.*Mark* = *Si*);
18: **if** ET_n.*Mark* == MO.*Mark* **then**
{
19: add ET_n ToBlockhain (BC, ET_{id}, *Si)*;
}
20: **else**
21: skip;
}

Based on the above algorithm, the entity requesting to join the private blockchain network have to send their unique identification to the main office. The main office will verify it and broadcast a message to the remaining entities in the network. By their turn, they have to verify and broadcast the message between each other until reaching consensus. And referring to the fault tolerance algorithm, if the number of broadcasted messages reached 2f+1, the main office will compute the received information including the nonce, message digest, and signature to create a special identification. If the special identification of the main office matches the special identification of the entity, it is concerned as an honest node, and the entity is added into the private blockchain. During the communication phase, the special identification will be used to authenticate the message and the sender.

3.2.2 Blockchain-based data communication

After verifying and registering the honest entities into the private blockchain, they can start communicating and sending information securely over the private blockchain network. Algorithm 2 depict the secure communication phase.

Algorithm 2. Communication phase

```
1: Input: Si of honest entities
2: Output: Secure communication
3: Process:
4: ET_s.Send(<Si, Msg, t, N>, request_M, ET_r);
// with request_M is the request to send a message, ET_s and ET_r are sender
  And receiver respectively
5: if BC_List.include (Si) and t < Δt = true then
 // Δt is the acceptable time delay
6:    {
7:        BC.convert (<CDHCK>, Msg, t, N);
8:        BC.encrypt (<NaCL>, Msg, t, N);
9:        New_message == BC.encrypted(<Msg, t, N>);
10:       BC.send (<New_message>, ET_r);
11:       ET_r.receive (<New_message>);
12:   }
13:  else
14:      skip;
     }
```

Based on the second algorithm, all messages are encrypted using the Curve25519 Diffie-Hellman cipher key (CFHCK). The generated message will be sent using NaCl Crypto Box algorithm which is highly efficient in time and secure [28]. Deploying this method, we can secure the communication between different authenticated entities is relatively short time period as it will be shown in the results.

3.3 Max-Cut Problem and QAOA Solution

Transportation in smart logistics forms a complex combinatorial optimization problem [29]. Deciding the most optimal path to take will adequately reduce the shipping and delivery time, which, by it turns, directly contribute to the reduction of carbon emission and cost. The combinatorial optimization problem is used to find the optimal object out of a set of defined objects. In terms of transportation in smart logistics, it can be used to find the heuristic optimal path in a large set of possibilities.

3.3.1 Max-cut problem formulation

The array of possibilities can be represented as a max-cut problem. Max-cut problem is one of the well-known NO-complete problems and can be applied in numerous fields [30], it can visually be represented as a graph where the more the graph nodes are, the higher possibility assignment we can get. To solve such a problem, Farhi et al. [31] proposed a hybrid QAOA. The goal of QAOA is to optimal parameters $(\boldsymbol{\beta_{opt}}, \boldsymbol{\gamma_{opt}})$ where $(\boldsymbol{\beta}, \boldsymbol{\gamma})$ are two parameters of the unitary gate $\boldsymbol{U}(\boldsymbol{\beta}, \boldsymbol{\gamma})$, to prepare the Quantum state $|\boldsymbol{\psi}(\boldsymbol{\beta}, \boldsymbol{\gamma})>$. Using this Quantum state, we can encode the solution to the problem as $|\boldsymbol{\psi}(\boldsymbol{\beta_{opt}}, \boldsymbol{\gamma_{opt}})>$.

The unitary $\boldsymbol{U}(\boldsymbol{\beta}, \boldsymbol{\gamma})$ is composed of two unitaries such as

$$\boldsymbol{U}(\boldsymbol{\beta}) = \boldsymbol{e}^{-i\boldsymbol{\beta}\boldsymbol{H_B}} \tag{1}$$

and

$$\boldsymbol{U}(\boldsymbol{\gamma}) = \boldsymbol{e}^{-i\boldsymbol{\gamma}\boldsymbol{H_B}} \tag{2}$$

where $\boldsymbol{H_B}$ refers to the mixing of Hamiltonian, and $\boldsymbol{H_P}$ refers to the problem Hamiltonian. Note that the Hamiltonian is the operator corresponding to the total energy of the Quantum system such as the Equation (3) and the energy of the system in state $|\boldsymbol{\psi}>$ given by expectation value can be described as in Equation

(4).

$$\boldsymbol{H} = \boldsymbol{H}^{\dagger} \tag{3}$$

$$\boldsymbol{E}(|\boldsymbol{\psi}>) = <\boldsymbol{\psi}|\boldsymbol{H}\,|\boldsymbol{\psi}> \tag{4}$$

The Quantum state can be prepared by applying the unitaries in form of blocks of both unitaries $\boldsymbol{U}(\boldsymbol{\beta},\boldsymbol{\gamma})$ for $\boldsymbol{p}$ times as described in Equation (5) where $|\boldsymbol{\psi_0}>$ refers to the suitable initial state.

$$|\boldsymbol{\psi}(\boldsymbol{\beta},\boldsymbol{\gamma})> = \boldsymbol{U}(\boldsymbol{\beta})\boldsymbol{U}(\boldsymbol{\gamma}) \ldots . \boldsymbol{U}(\boldsymbol{\beta_p})\boldsymbol{U}(\boldsymbol{\gamma_p})\,|\boldsymbol{\psi_0}> \tag{5}$$

Fig. 2 demonstrates the max-cut problem discussed above; the implementation of this problem, circuits construction, and simulation was conducted using IBM Quantum Lab. In this scenario, the points in the graph represent the pit stops that the transportation in smart logistics has to make from the supplier to the end client.

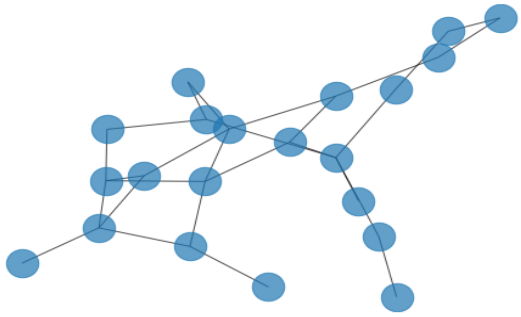


Fig. 2. Visual interpretation of max-cut problem.

We can represent the max-cut using the Hamiltonian problem such as the Equation (6).

$$\begin{aligned}\boldsymbol{H_p} = \frac{\boldsymbol{1}}{\boldsymbol{2}}(\boldsymbol{Z_0}\otimes\boldsymbol{Z_1}\otimes\boldsymbol{I_2}\otimes\boldsymbol{I_3}) + \frac{\boldsymbol{1}}{\boldsymbol{2}}(\boldsymbol{I_0}\otimes\boldsymbol{Z_1}\otimes\boldsymbol{Z_2}\otimes\boldsymbol{I_3}) + \frac{\boldsymbol{1}}{\boldsymbol{2}}(\boldsymbol{Z_0}\otimes\boldsymbol{I_1}\otimes\boldsymbol{I_2}\otimes\boldsymbol{Z_3}) \\ + \frac{\boldsymbol{1}}{\boldsymbol{2}}(\boldsymbol{I_0}\otimes\boldsymbol{I_1}\otimes\boldsymbol{Z_2}\otimes\boldsymbol{Z_3})\end{aligned} \tag{6}$$

3.3.2 QAOA-based solution

For the above descripted problem, QAOA solution can be implemented. Equations (7), (8), and (9) depicts the necessary steps to follow to build the solution for (6).

The mixer Hamiltonian $\boldsymbol{H_B}$ can be described as follow:

$$\begin{aligned}\boldsymbol{H_B} = (\boldsymbol{X_0}\otimes\boldsymbol{I_1}\otimes\boldsymbol{I_2}\otimes\boldsymbol{I_3}) + (\boldsymbol{I_0}\otimes\boldsymbol{X_1}\otimes\boldsymbol{I_2}\otimes\boldsymbol{I_3}) + (\boldsymbol{I_0}\otimes\boldsymbol{I_1}\otimes\boldsymbol{X_2}\otimes\boldsymbol{I_3}) \\ + (\boldsymbol{I_0}\otimes\boldsymbol{I_1}\otimes\boldsymbol{I_2}\otimes\boldsymbol{X_3})\end{aligned} \tag{7}$$

From the equations above, the unitary $\boldsymbol{U}$ corresponding to $\boldsymbol{H_B}$ and $\boldsymbol{H_p}$ can be represented as follows, where each product is corresponding to an X rotation on a Qubit:

$$\boldsymbol{U}(\boldsymbol{H_B}) = \boldsymbol{e}^{-i\beta\boldsymbol{H_B}} = \boldsymbol{e}^{-i\beta\boldsymbol{X_0}}\boldsymbol{e}^{-i\beta\boldsymbol{X_1}}\boldsymbol{e}^{-i\beta\boldsymbol{X_2}}\,\boldsymbol{e}^{-i\beta\boldsymbol{X_3}} \tag{8}$$

$$\boldsymbol{U}(\boldsymbol{H_P}) = \boldsymbol{e}^{-i\gamma\boldsymbol{H_P}} = \boldsymbol{e}^{-i\gamma\boldsymbol{Z_0Z_1}}\boldsymbol{e}^{-i\gamma\boldsymbol{Z_1Z_2}}\boldsymbol{e}^{-i\gamma\boldsymbol{Z_2Z_3}}\,\boldsymbol{e}^{-i\gamma\boldsymbol{Z_0Z_3}} \tag{9}$$

The representative Quantum circuits are described in Fig. 3.

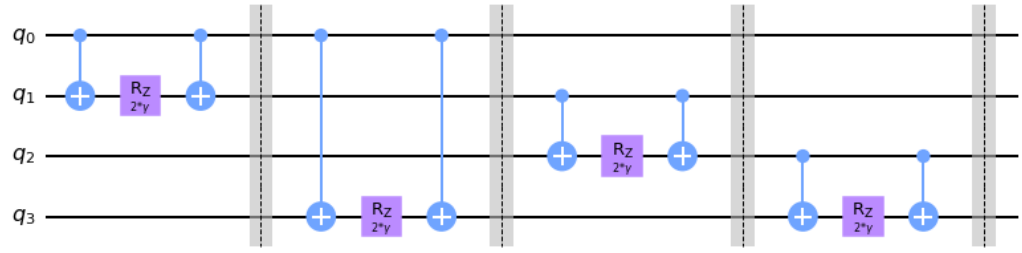


Fig. 3. Quantum circuit for unitary problem.

In the above, we discussed the problem formulation and preparation of the Quantum state to initialize a QAOA. First, we initialized the states, we applied the unitary $\boldsymbol{U(H_P)} = \boldsymbol{e^{-i\gamma H_P}}$ and then we applied the mixing unitary $\boldsymbol{U(H_B)} = \boldsymbol{e^{-i\beta H_B}}$. Now, we will implement these steps to build the representative Quantum circuit for QAOA as depicted in Fig. 4.

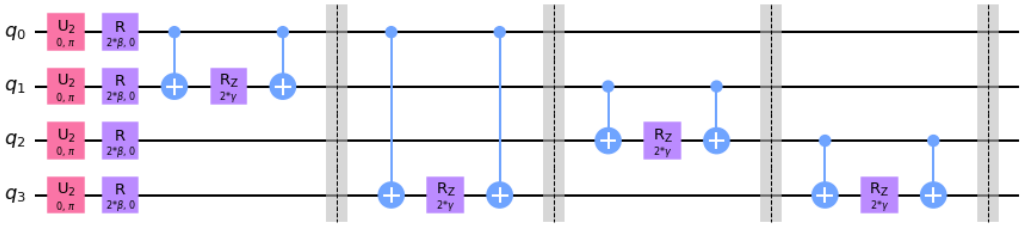


Fig. 4. Quantum circuit for QAOA.

Based on the QAOA circuit, we can find the optimal parameters $(\boldsymbol{\beta_{opt}, \gamma_{opt}})$ with the expectation value such as (10) is minimized. Measuring Z-basis allows returning such expectations.

$$< \boldsymbol{\psi(\beta_{opt}, \gamma_{opt})}| \boldsymbol{H_P}|\boldsymbol{\psi(\beta_{opt}, \gamma_{opt})} > \tag{10}$$

4. Performance Evaluation

All of the above-mentioned simulations have been conducted using IBM Quantum Lab and Qiskit V0.32.0. Fig. 5 presents the version information related to the Qiskit software package and system information of the simulation environment. IBM Hyperledger was deployed for the blockchain-based secure communication simulation on a Linux OS, and 8 CPUs.

Finally, a classical optimization algorithm is used to find the optimal parameters. The algorithm can be implemented in two main steps.

Firstly, the initialization of $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ to real values, and then we repeat the initialization phase until the suitable convergence criteria are met. For simulation purposes, the COBYLA algorithm is deployed in IBM Qiskit. The results are depicted in Fig. 6 show the successful optimization of the above-discussed problem.

While Fig. 7 presents the probabilities of the heuristic optimized path; the gates with higher probabilities represent the best heuristic optimized path for the transportation in a smart logistic system take.

Version Information

Qiskit Software	Version
qiskit-terra	0.18.3
qiskit-aer	0.9.1
qiskit-ignis	0.6.0
qiskit-ibmq-provider	0.18.0
qiskit-aqua	0.9.5
qiskit	0.32.0
qiskit-nature	0.2.2
qiskit-finance	0.2.1
qiskit-optimization	0.2.3
qiskit-machine-learning	0.2.1
System information	
Python	3.8.10 | packaged by conda-forge | (default, May 11 2021, 07:01:05) [GCC 9.3.0]
OS	Linux
CPUs	8
Memory (Gb)	31.408958435058594
	Wed Nov 17 08:27:11 2021 UTC

Fig. 5. Simulation environment.

```
[47]: from scipy.optimize import minimize


      expectation = get_expectation(G, p=100)

      res = minimize(expectation,
                           [1.0, 1.0],
                           method='COBYLA')
      res
```

```
[47]:      fun: -2.994140625
         maxcv: 0.0
       message: 'Optimization terminated successfully.'
          nfev: 30
        status: 1
       success: True
             x: array([1.9793337 , 1.16663483])
```

Fig. 6. Successful optimization results.

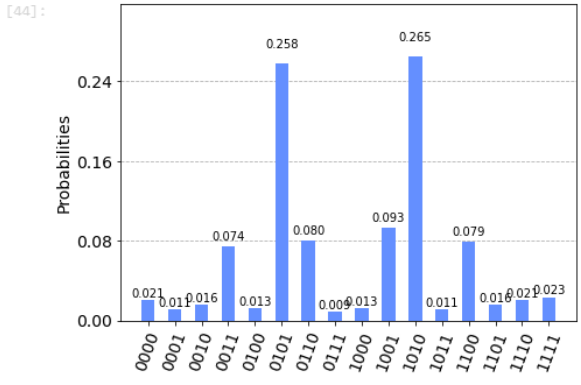


Fig. 7. Probabilities of the heuristic optimized path.

Table 1 depict the performance of the proposed blockchain-based QAOA framework for logistics. Implementing this framework on a case-scenario environment shows a high successful route optimization rate by a 91%, calculating this results on the simulated case scenario shows a possible cost reduction of 10%, this is due to the time reduction and used fuel which reduce the carbon emission by 6%. Using private blockchain for communication will consume 8.3 seconds in total, however, since the communication between different tiers of smart factory and logistics system is latency-tolerant, these results are proven to be heuristic and highly secure. As our future work, we plan to reduce the communication time in order to create a more efficient message transmission system while maintaining the security and privacy of the transmitted messages.

Table 1. Performance of blockchain-based QAOA

Task	Unit	Performance
Blockchain-based registration	s	2.3
Blockchain-based message transmission	s	5
Route optimization		
Computing	s	8.6
Decision	ms	5
Successful optimized route probability	%	91
Cost reduction	%	10
Carbon footprint reduction	%	6
System scalability improvement	%	23

5. Conclusion

A QAOA, as a variational Quantum algorithm for approximately solving discrete combinatorial optimization problems, can be deployed into the smart logistics dilemma to improve the scalability of the system, decrease the time, thus reducing the carbon footprint and smart manufacturing system cost. Moreover, Blockchain, as a secure distributed ledger, is capable of bringing the desired security to the smart logistic system. In this paper, we proposed the deployment of both QAOA and blockchain for a secure heuristic optimization in smart logistics. The results are promising to improve the scalability of the system by 23%, reduce the carbon footprint by 6%, and cut down the transportation and logistics costs by 10%.

Acknowledgments

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Author’s Contributions

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Competing Interests

The authors declare that they have no competing interests.

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